ELITE IGCSE ACADEMY

Private Past Paper Solutions

May 2023 P1H Worked Solutions

May 2023 | P1H

27 questions ▪ question image followed by worked solution

PREPARED BY

Dr. Eslam Ahmed

Assistant Lecturer, Cairo University Faculty of Engineering

WhatsApp: +20 112 000 9622 | eliteigcse.com

TEACHER EDITION

Worked answers included

PRIVATE

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PAST PAPER QUESTION

Q#: 1 Marks: 3

TOPIC: **Ratio Toolkit**

May 2023 P1H - May 2023 | P1H

1 Last season, the number of goals scored by Arjun, by Simon and by Kath for their football team were in the ratios $2:5:8$

Simon scored 12 more goals than Arjun.

Work out the number of goals scored by Kath.

....

(Total for Question 1 is 3 marks)

PAST PAPER SOLUTION

Q#: 1 Marks: 3

TOPIC: **Ratio Toolkit**

May 2023 P1H - May 2023 | P1H

WORKED SOLUTION

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Topic check. Ratio Toolkit. The tag is correct.

Method

The ratio is

$$\text{Arjun} : \text{Simon} : \text{Kath} = 2 : 5 : 8$$

Simon scored 12 more than Arjun.

$$5 - 2 = 3 \text{ parts}$$

$$3 \text{ parts} = 12$$

$$1 \text{ part} = 4$$

Kath scored

$$8 \times 4 = 32$$

FINAL ANSWER

32

PAST PAPER QUESTION

Topic: **Statistics Toolkit**

May 2023 P1H - May 2023 | P1H

Q#: 2 Marks: 3

2 The table gives information about the number of minutes that Abby spent walking each day in September.

Number of minutes (M)	**Frequency**
$0 < M \leqslant 30$	5
$30 < M \leqslant 60$	6
$60 < M \leqslant 90$	8
$90 < M \leqslant 120$	9
$120 < M \leqslant 150$	2

Work out an estimate for the total number of minutes that Abby spent walking in September.

.... minutes

(Total for Question 2 is 3 marks)

PAST PAPER SOLUTION

Q#: 2 Marks: 3

TOPIC: **Statistics Toolkit**

May 2023 P1H - May 2023 | P1H

WORKED SOLUTION

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Topic check. Statistics Toolkit. The tag is correct.

Solution

Use midpoints $15, 45, 75, 105, 135$.

$$15(5) + 45(6) + 75(8) + 105(9) + 135(2) = 2160$$

FINAL ANSWER

2160 minutes.

PAST PAPER QUESTION

Q#: 3 Marks: 4

TOPIC: **Percentages**

May 2023 P1H - May 2023 | P1H

3 Nanette buys 60 notebooks for a total cost of 780 dirhams.

Nanette sells 70% of the notebooks for 22 dirhams each.
She sells the remaining notebooks for 19 dirhams each.

Work out Nanette's percentage profit.
Give your answer correct to 3 significant figures.

....%

(Total for Question 3 is 4 marks)

PAST PAPER SOLUTION

TOPIC: **Percentages**

Q#: 3 Marks: 4

May 2023 P1H - May 2023 | P1H

WORKED SOLUTION

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Topic check. Percentages. The tag is correct.

Method

Number sold for 22 dirhams:

$$70\% \times 60 = 42$$

Number remaining:

$$60 - 42 = 18$$

Total selling price:

$$42 \times 22 + 18 \times 19 = 924 + 342 = 1266$$

Profit:

$$1266 - 780 = 486$$

Percentage profit:

$$\frac{486}{780} \times 100 = 62.307\ldots\%$$

FINAL ANSWER

62.3%

PAST PAPER QUESTION

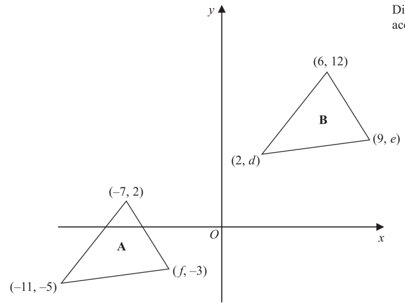
Q#: 4 **Marks: 8**

Topic: **Transformations**

May 2023 P1H - May 2023 | P1H

4 The diagram shows a sketch of triangle **A** and triangle **B** on a coordinate grid.

Diagram **NOT** accurately drawn



(a) Given that triangle **A** has been translated to give triangle **B**, work out the value of d, the value of e and the value of f

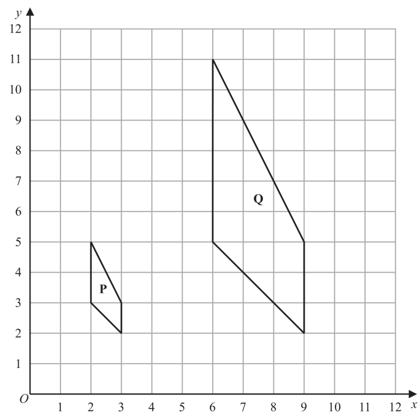
d = ...

e = ...

f = ...

(3)

The diagram shows shape **P** and shape **Q** drawn on a grid.



(b) Describe fully the single transformation that maps shape **P** onto shape **Q**

....

....

(3)

(c) On the grid above, rotate shape **P** 90° clockwise about (3, 5)
Label your shape **R**

(2)

(Total for Question 4 is 8 marks)

PAST PAPER SOLUTION

Q#: 4 Marks: 8

TOPIC: **Transformations**

May 2023 P1H - May 2023 | P1H

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Topic check. Transformations. The tag is correct.

Solution

(a) The translation from $(-7, 2)$ to $(6, 12)$ is

$$\begin{pmatrix} 13 \\ 10 \end{pmatrix}$$

So

$$(-11, -5) \mapsto (2, 5)$$

and

$$(f, -3) \mapsto (9, 7)$$

Therefore

$$d = 5, \qquad e = 7, \qquad f = -4$$

(b) Shape P maps to Q by an enlargement with scale factor 3 and centre $(0, 2)$.
(c) Rotate P 90° clockwise about $(3, 5)$. The image has vertices

$$(3, 6),\ (1, 6),\ (0, 5),\ (1, 5)$$

FINAL ANSWER

(a) $d = 5$, $e = 7$, $f = -4$; (b) enlargement, scale factor 3, centre $(0, 2)$; (c) vertices $(3, 6), (1, 6), (0, 5), (1, 5)$.

PAST PAPER QUESTION

Q#: 5 Marks: 4

TOPIC: **Right-Angled Triangles - Pythagoras & Trigonometry**

May 2023 P1H - May 2023 | P1H

5 The diagram shows a shaded shape *AEBCD* made by removing triangle *AEB* from rectangle *ABCD*

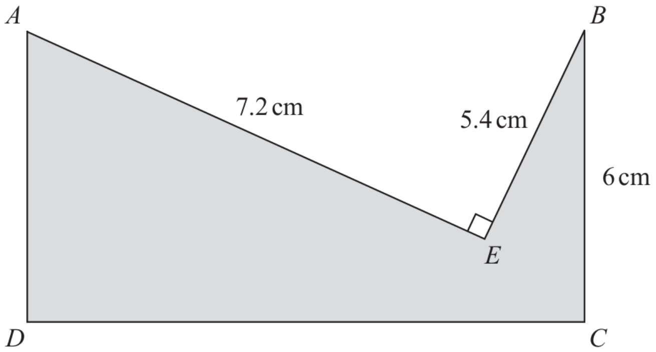

Diagram **NOT** accurately drawn

$AE = 7.2$ cm $BE = 5.4$ cm $BC = 6$ cm angle $AEB = 90°$

Work out the perimeter of the shaded shape.

.... cm

(Total for Question 5 is 4 marks)

PAST PAPER SOLUTION

Q#: 5 Marks: 4

TOPIC: **Right-Angled Triangles - Pythagoras & Trigonometry**

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Topic check. Right-Angled Triangles - Pythagoras & Trigonometry. The tag is correct.

Solution

Triangle AEB is right-angled:

$$AB^2 = 7.2^2 + 5.4^2 = 81$$

$$AB = 9$$

For rectangle $ABCD$,

$$DC = 9, \qquad AD = BC = 6$$

Perimeter of the shaded shape:

$$7.2 + 5.4 + 6 + 9 + 6 = 33.6$$

FINAL ANSWER

33.6 cm.

PAST PAPER QUESTION

Q#: 6 Marks: 8

Topic: **Factorising**

May 2023 P1H - May 2023 | P1H

6 (a) Simplify $(2c^4d^7)^3$

....

(2)

(b) Find the value of $5y^0$ where $y > 0$

....

(1)

(c) Factorise fully $16a^2b^3 + 20a^3b$

....

(2)

(d) (i) Factorise $x^2 + 9x - 22$

....

(2)

(ii) Hence solve $x^2 + 9x - 22 = 0$

....

(1)

(Total for Question 6 is 8 marks)

PAST PAPER SOLUTION

Q#: 6 Marks: 8

TOPIC: **Factorising**

May 2023 P1H - May 2023 | P1H

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Topic check. Factorising. The tag is correct.

Solution

(a)

$$(2c^4d^7)^3 = 8c^{12}d^{21}$$

(b) $5y^0 = 5$, since $y^0 = 1$.

(c)

$$16a^2b^3 + 20a^3b = 4a^2b(4b^2 + 5a)$$

(d)(i)

$$x^2 + 9x - 22 = (x + 11)(x - 2)$$

(d)(ii) $x = -11$ or $x = 2$.

FINAL ANSWER

$8c^{12}d^{21}$, 5, $4a^2b(4b^2 + 5a)$, $(x + 11)(x - 2)$, $x = -11$ or $x = 2$

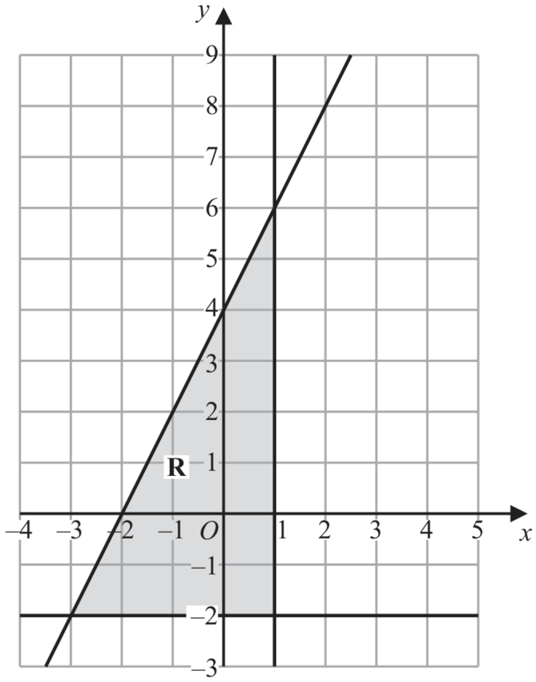
PAST PAPER QUESTION

Q#: 7 Marks: 4

Topic: **Graphing Inequalities**

May 2023 P1H - May 2023 | P1H

7



The region **R**, shown shaded in the diagram, is bounded by three straight lines.

Find the inequalities that define **R**

....

....

....

(Total for Question 7 is 4 marks)

PAST PAPER SOLUTION

Q#: 7 Marks: 4

TOPIC: **Graphing Inequalities**

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Topic check. Correct. This is identifying inequalities from a shaded region.

Solution

The vertical boundary is

$$x = 1$$

so the region has

$$x \leq 1$$

The horizontal boundary is

$$y = -2$$

so

$$y \geq -2$$

The sloping line passes through $(0, 4)$ and has gradient 2, so

$$y = 2x + 4$$

The shaded region is below this line:

$$y \leq 2x + 4$$

FINAL ANSWER

$x \leq 1$, $y \geq -2$, $y \leq 2x + 4$.

PAST PAPER QUESTION

Q#: 8 Marks: 4

TOPIC: **Prime Factors, HCF & LCM**

May 2023 P1H - May 2023 | P1H

8 (a) Write 300 as a product of its prime factors.
Show your working clearly.

....

(2)

$A = 2 \times 2 \times 2 \times 3 \times 3 \times 5$

$B = 2 \times 2 \times 3 \times 3 \times 3 \times 5$

(b) Find the lowest common multiple (LCM) of $5A$ and $7B$
Show your working clearly.

....

(2)

(Total for Question 8 is 4 marks)

PAST PAPER SOLUTION

Q#: 8 Marks: 4

TOPIC: **Prime Factors, HCF & LCM**

May 2023 P1H - May 2023 | P1H

WORKED SOLUTION

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Topic check. Prime factors, HCF and LCM. The tag is correct.

Method

First write 300 as a product of prime factors:

$$300 = 3 \times 100 = 3 \times 2^2 \times 5^2$$

$$300 = 2^2 \times 3 \times 5^2$$

Now

$$A = 2^3 \times 3^2 \times 5$$

and

$$B = 2^2 \times 3^3 \times 5$$

So

$$5A = 2^3 \times 3^2 \times 5^2$$

and

$$7B = 2^2 \times 3^3 \times 5 \times 7$$

For the LCM of $5A$ and $7B$, take the highest power of every prime:

$$\text{LCM} = 2^3 \times 3^3 \times 5^2 \times 7$$

FINAL ANSWER

(a) $2^2 \times 3 \times 5^2$, (b) $2^3 \times 3^3 \times 5^2 \times 7$

PAST PAPER QUESTION

TOPIC: **Simultaneous Equations**

May 2023 P1H - May 2023 | P1H

Q#: 9 Marks: 3

9 Solve the simultaneous equations

$$2x + 9y = 14.5$$
$$7x + 3y = 8$$

Show clear algebraic working.

$x =$ ....

$y =$ ....

(Total for Question 9 is 3 marks)

PAST PAPER SOLUTION

Q#: 9 Marks: 3

TOPIC: **Simultaneous Equations**

May 2023 P1H - May 2023 | P1H

WORKED SOLUTION

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Topic check. Simultaneous equations. The tag is correct.

Solution

$$2x + 9y = 14.5$$

$$7x + 3y = 8$$

Multiply the second equation by 3:

$$21x + 9y = 24$$

Subtract the first equation:

$$19x = 9.5$$

$$x = 0.5$$

Substitute into $7x + 3y = 8$:

$$7(0.5) + 3y = 8$$

$$3.5 + 3y = 8$$

$$y = 1.5$$

FINAL ANSWER

$x = 0.5$, $y = 1.5$.

PAST PAPER QUESTION

Q#: 10 Marks: 2

TOPIC: **Statistics Toolkit**

May 2023 P1H - May 2023 | P1H

10 Here are the test marks of 15 students.

7 10 14 15 16 17 18 19 20 20 23 25 30 39 40

Find the interquartile range of these marks.

....

(Total for Question 10 is 2 marks)

PAST PAPER SOLUTION

Q#: 10 Marks: 2

TOPIC: **Statistics Toolkit**

May 2023 P1H - May 2023 | P1H

WORKED SOLUTION

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Topic check. Statistics Toolkit. The tag is correct.

Solution

The marks in order are

$$7,\ 10,\ 14,\ 15,\ 16,\ 17,\ 18,\ 19,\ 20,\ 20,\ 23,\ 25,\ 30,\ 39,\ 40$$

$$Q_1 = 15, \qquad Q_3 = 25$$

$$\text{IQR} = 25 - 15 = 10$$

FINAL ANSWER

10.

PAST PAPER QUESTION

Q#: 11 Marks: 6

TOPIC: **Differentiation**

May 2023 P1H - May 2023 | P1H

11 The curve **C** has equation $y = 4x^3 + x^2 - 20x$

(a) Find $\dfrac{dy}{dx}$

$\dfrac{dy}{dx} =$ ....

(2)

(b) Find the x coordinates of the points on **C** where the gradient is 4
Show clear algebraic working.

....

(4)

(Total for Question 11 is 6 marks)

PAST PAPER SOLUTION Q#: 11 Marks: 6

TOPIC: **Differentiation**

May 2023 P1H - May 2023 | P1H

WORKED SOLUTION

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Topic check. Corrected to differentiation. The question is about derivatives and gradients of a curve.

Solution

$$y = 4x^3 + x^2 - 20x$$

(a)

$$\frac{dy}{dx} = 12x^2 + 2x - 20$$

(b) The gradient is 4, so

$$12x^2 + 2x - 20 = 4$$

$$12x^2 + 2x - 24 = 0$$

$$6x^2 + x - 12 = 0$$

Factorise:

$$(3x - 4)(2x + 3) = 0$$

So

$$x = \frac{4}{3} \quad \text{or} \quad x = -\frac{3}{2}$$

FINAL ANSWER

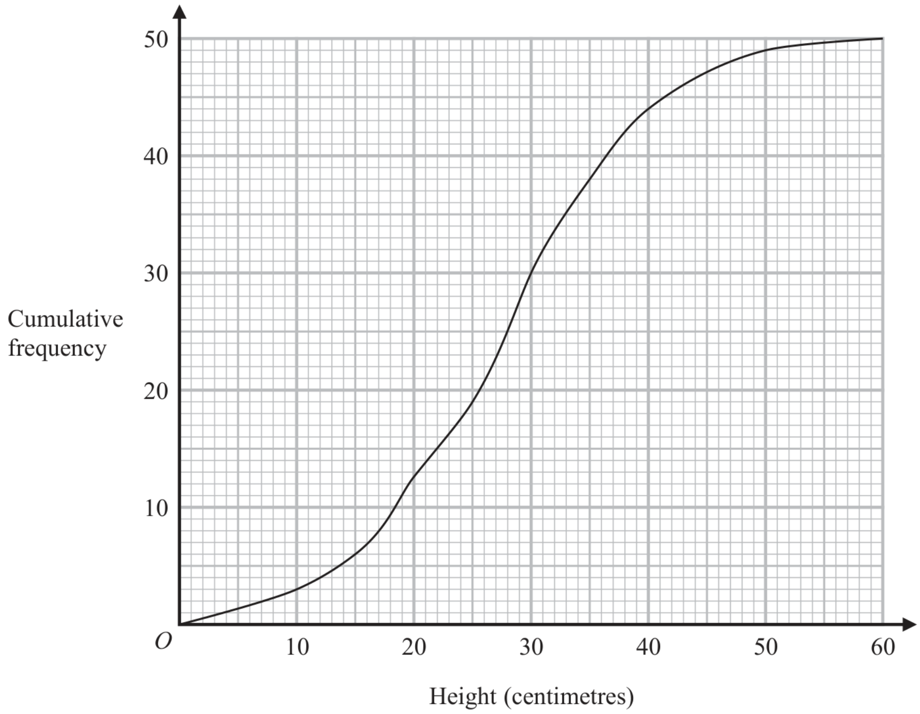
$\frac{dy}{dx} = 12x^2 + 2x - 20$, and $x = \frac{4}{3}$ or $x = -\frac{3}{2}$.

PAST PAPER QUESTION Q#: 12 Marks: 4

TOPIC: **Cumulative Frequency Diagrams**

May 2023 P1H - May 2023 | P1H

12 The cumulative frequency graph shows information about the heights, in centimetres, of 50 plants in a flowerbed.



(a) Use the graph to find an estimate for the median height of these plants.

.... centimetres
(1)

(b) Use the graph to find the frequency for the class interval $30 < \text{Height} \leqslant 40$

....
(1)

(c) Use the graph to find an estimate for the number of plants with a height greater than 35 centimetres.

....
(2)

(Total for Question 12 is 4 marks)

PAST PAPER SOLUTION

Q#: 12 Marks: 4

TOPIC: **Cumulative Frequency Diagrams**

May 2023 P1H - May 2023 | P1H

WORKED SOLUTION

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Topic check. Cumulative Frequency Diagrams. The tag is correct.

Solution

There are 50 plants, so the median is the 25th value.
From the cumulative frequency graph,

$$\text{median} \approx 27 \text{ cm}$$

For the interval $30 < h \leq 40$,

$$F(40) - F(30) \approx 44 - 31 = 13$$

For plants taller than 35 cm,

$$50 - F(35) \approx 50 - 37 = 13$$

FINAL ANSWER

median about 27 cm, about 13 plants, about 13 plants.

PAST PAPER QUESTION

Q#: 13 Marks: 3

TOPIC: **Coordinate Geometry**

May 2023 P1H - May 2023 | P1H

13 *A* is the point with coordinates (–5, 12)

B is the point with coordinates (19, –48)

Find an equation of the straight line that passes through the points *A* and *B*

....

(Total for Question 13 is 3 marks)

PAST PAPER SOLUTION

Q#: 13 Marks: 3

TOPIC: **Coordinate Geometry**

May 2023 P1H - May 2023 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Coordinate geometry. The tag is correct.

Solution

$$A = (-5, 12), \qquad B = (19, -48)$$

The gradient is

$$\frac{-48 - 12}{19 - (-5)} = \frac{-60}{24} = -\frac{5}{2}$$

Using point $(-5, 12)$:

$$y - 12 = -\frac{5}{2}(x + 5)$$

Multiply by 2:

$$2y - 24 = -5x - 25$$

$$5x + 2y + 1 = 0$$

FINAL ANSWER
$5x + 2y + 1 = 0$.

PAST PAPER QUESTION

Q#: 14 Marks: 3

TOPIC: **Factorising**

May 2023 P1H - May 2023 | P1H

14 Factorise fully $50g^2 - 18$

....

(Total for Question 14 is 3 marks)

PAST PAPER SOLUTION

Q#: 14 Marks: 3

TOPIC: **Factorising**

May 2023 P1H - May 2023 | P1H

WORKED SOLUTION

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Topic check. Factorising. The tag is correct.

Solution

$$50g^2 - 18 = 2(25g^2 - 9)$$

$$= 2(5g - 3)(5g + 3)$$

FINAL ANSWER

$2(5g - 3)(5g + 3)$

PAST PAPER QUESTION Q#: 15 Marks: 6

TOPIC: **Powers, Roots & Standard Form**

May 2023 P1H - May 2023 | P1H

15 (a) $\sqrt{2} \div \dfrac{8^3}{16^{\frac{3}{2}}} = 2^n$

Work out the value of n

Show your working clearly.

$n =$ ....

(3)

(b) Find 4% of 4.5×10^{157}

Give your answer in standard form.

....

(3)

(Total for Question 15 is 6 marks)

PAST PAPER SOLUTION

Q#: 15 Marks: 6

TOPIC: **Powers, Roots & Standard Form**

May 2023 P1H - May 2023 | P1H

WORKED SOLUTION

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Topic check. Powers, roots and standard form. The tag is correct.

Method

$$\sqrt{2} = 2^{\frac{1}{2}}$$

$$8^3 = (2^3)^3 = 2^9$$

$$16^{\frac{3}{2}} = (2^4)^{\frac{3}{2}} = 2^6$$

So

$$\frac{8^3}{16^{\frac{3}{2}}} = \frac{2^9}{2^6} = 2^3$$

Therefore

$$\sqrt{2} \div \frac{8^3}{16^{\frac{3}{2}}} = 2^{\frac{1}{2}} \div 2^3$$

$$= 2^{\frac{1}{2}-3} = 2^{-\frac{5}{2}}$$

So

$$n = -\frac{5}{2}$$

For part (b),

$$4\% = 0.04 = 4 \times 10^{-2}$$

$$0.04(4.5 \times 10^{157}) = 18 \times 10^{155}$$

$$= 1.8 \times 10^{156}$$

FINAL ANSWER

(a) $n = -\frac{5}{2}$, (b) 1.8×10^{156}

PAST PAPER QUESTION

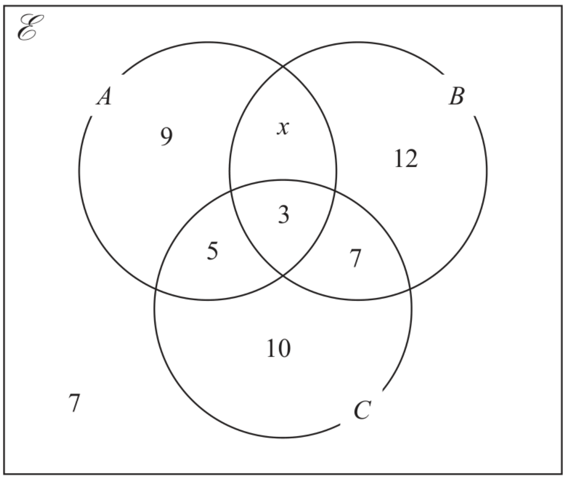
Q#: 16 Marks: 3

TOPIC: **Set Notation & Venn Diagrams**

May 2023 P1H - May 2023 | P1H

16 The Venn diagram shows a universal set $\mathscr{E}$ and sets A, B and C

The numbers and the letter x represent **numbers** of elements.



Given that $n(A \cup B) = 42$

(a) find the value of x

$x =$ ....

(1)

(b) Find $n(A')$

....

(1)

(c) Find $n(B' \cap C)$

....

(1)

(Total for Question 16 is 3 marks)

PAST PAPER SOLUTION Q#: 16 Marks: 3

TOPIC: **Set Notation & Venn Diagrams**

May 2023 P1H - May 2023 | P1H

WORKED SOLUTION

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Topic check. Set notation and Venn diagrams. The tag is correct.

Method

The regions inside $A \cup B$ are

$$9 + x + 12 + 5 + 3 + 7$$

We are told that

$$n(A \cup B) = 42$$

So

$$36 + x = 42$$

$$x = 6$$

For A', add every region not inside A:

$$12 + 7 + 10 + 7 = 36$$

For $B' \cap C$, add the regions that are in C but not in B:

$$5 + 10 = 15$$

FINAL ANSWER

(a) $x = 6$, (b) 36, (c) 15

PAST PAPER QUESTION **Q#: 17** **Marks: 4**

TOPIC: **Functions**

May 2023 P1H - May 2023 | P1H

17 The functions g and h are such that

$$g(x) = \frac{11}{2x - 5}$$

$$h(x) = x^2 + 4 \qquad x \geqslant 0$$

(a) What value of x must be excluded from any domain of g?

....

(1)

(b) Solve gh(x) = 1

....

(3)

(Total for Question 17 is 4 marks)

PAST PAPER SOLUTION

Q#: 17 Marks: 4

TOPIC: **Functions**

May 2023 P1H - May 2023 | P1H

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Topic check. Corrected to Functions. The question tests domain and composite functions.

Solution

$$g(x) = \frac{11}{2x-5}, \qquad h(x) = x^2 + 4, \quad x \geq 0$$

(a) The denominator of $g(x)$ cannot be zero.

$$2x - 5 = 0$$

$$x = \frac{5}{2}$$

So $x = \frac{5}{2}$ must be excluded.
(b)

$$gh(x) = g(h(x)) = \frac{11}{2(x^2+4)-5}$$

$$= \frac{11}{2x^2+3}$$

Solve

$$\frac{11}{2x^2+3} = 1$$

$$2x^2 + 3 = 11$$

$$2x^2 = 8$$

$$x^2 = 4$$

Since $x \geq 0$,

$$x = 2$$

FINAL ANSWER
Exclude $x = \frac{5}{2}$, and $x = 2$.

PAST PAPER QUESTION

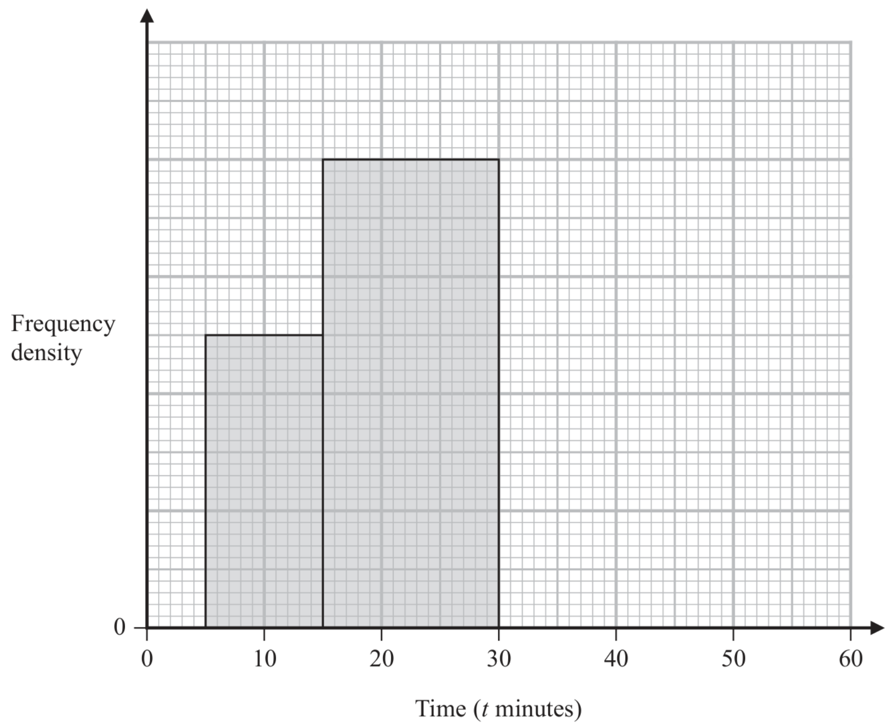
Q#: 18 Marks: 4

TOPIC: **Histograms**

May 2023 P1H - May 2023 | P1H

18 The incomplete table and incomplete histogram give information about the times, in minutes, that 140 people waited at a station for a train.

Time (*t* minutes)	**Frequency**
$0 < t \leqslant 5$	23
$5 < t \leqslant 15$	
$15 < t \leqslant 30$	
$30 < t \leqslant 40$	18
$40 < t \leqslant 60$	14



Complete the table and the histogram.

(Total for Question 18 is 4 marks)

PAST PAPER SOLUTION

Q#: 18 Marks: 4

TOPIC: **Histograms**

May 2023 P1H - May 2023 | P1H

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Topic check. Histograms. The tag is correct.

Solution

The total frequency is 140.
The known frequencies are

$$23 + 18 + 14 = 55$$

So the two missing frequencies have total

$$140 - 55 = 85$$

From the histogram, the missing frequencies are

$$5 < t \leq 15: \ 25, \qquad 15 < t \leq 30: \ 60$$

The missing histogram densities are

$$0 < t \leq 5 : \frac{23}{5} = 4.6$$

$$30 < t \leq 40 : \frac{18}{10} = 1.8$$

$$40 < t \leq 60 : \frac{14}{20} = 0.7$$

FINAL ANSWER

missing frequencies $25, 60$, and missing densities $4.6, 1.8, 0.7$.

PAST PAPER QUESTION

Q#: 19 Marks: 5

TOPIC: **Vectors**

May 2023 P1H - May 2023 | P1H

19

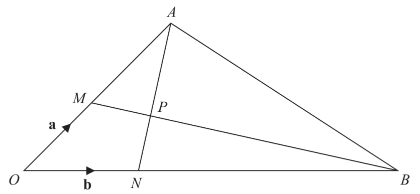

Diagram **NOT** accurately drawn

OMA, *ONB*, *MPB* and *NPA* are straight lines.

M is the midpoint of *OA*

$ON : NB = 1 : 5$

$\overrightarrow{OM} = \mathbf{a}$ $\overrightarrow{ON} = \mathbf{b}$

(a) Find in terms of **a** and **b** the vector $\overrightarrow{AN}$

....

(1)

(b) Use a vector method to find the ratio $AP : PN$

$AP : PN =$ ....

(4)

(Total for Question 19 is 5 marks)

PAST PAPER SOLUTION

Q#: 19 Marks: 5

TOPIC: **Vectors**

May 2023 P1H - May 2023 | P1H

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Topic check. Corrected to Vectors. The ratio is found using a vector method.

Method

Since M is the midpoint of OA,

$$\overrightarrow{OA} = 2a$$

Also,

$$\overrightarrow{ON} = b$$

So

$$\overrightarrow{AN} = \overrightarrow{ON} - \overrightarrow{OA} = b - 2a$$

Let

$$\overrightarrow{AP} = \lambda\overrightarrow{AN}$$

Then

$$\overrightarrow{OP} = 2a + \lambda(b - 2a) = (2 - 2\lambda)a + \lambda b$$

Since $ON : NB = 1 : 5$,

$$\overrightarrow{OB} = 6b$$

Point P is also on MB, so

$$\overrightarrow{OP} = a + \mu(6b - a) = (1 - \mu)a + 6\mu b$$

Compare coefficients:

$$2 - 2\lambda = 1 - \mu$$

$$\lambda = 6\mu$$

From $\mu = 2\lambda - 1$,

$$\lambda = 6(2\lambda - 1)$$

$$11\lambda = 6$$

$$\lambda = \frac{6}{11}$$

So

$$AP : PN = \frac{6}{11} : \frac{5}{11} = 6 : 5$$

FINAL ANSWER

(a) $b - 2a$, (b) $6 : 5$

PAST PAPER QUESTION

Q#: 20 Marks: 6

TOPIC: **Sequences**

May 2023 P1H - May 2023 | P1H

20 The sum of the first 80 terms of an arithmetic series, S, is 470

The 75th term of S is 14.5

The sum of the first X terms of S is 171

Work out the value of X
Show your working clearly.

$X =$ ....

(Total for Question 20 is 6 marks)

PAST PAPER SOLUTION

Q#: 20 Marks: 6

TOPIC: **Sequences**

May 2023 P1H - May 2023 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

$$S_{80} = 470$$

$$40(2a + 79d) = 470$$

$$2a + 79d = \frac{47}{4}$$

The 75th term is

$$a + 74d = 14.5 = \frac{29}{2}$$

Double it:

$$2a + 148d = 29$$

Subtract:

$$69d = 29 - \frac{47}{4} = \frac{69}{4}$$

$$d = \frac{1}{4}$$

$$a + 74\left(\frac{1}{4}\right) = \frac{29}{2}$$

$$a = -4$$

Now

$$S_X = 171$$

$$171 = \frac{X}{2}\left(2(-4) + (X-1)\frac{1}{4}\right)$$

$$171 = \frac{X(X-33)}{8}$$

$$X^2 - 33X - 1368 = 0$$

$$X = \frac{33+81}{2} = 57$$

FINAL ANSWER

$X = 57$

PAST PAPER QUESTION

Q#: 20 Marks: 6

TOPIC: **Sequences**

May 2023 P1H - May 2023 | P1H

20 The sum of the first 80 terms of an arithmetic series, S, is 470

The 75th term of S is 14.5

The sum of the first X terms of S is 171

Work out the value of X
Show your working clearly.

$X =$ ....

(Total for Question 20 is 6 marks)

PAST PAPER SOLUTION

Q#: 20 Marks: 6

TOPIC: **Sequences**

May 2023 P1H - May 2023 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

$$S_{80} = 470$$

$$40(2a + 79d) = 470$$

$$2a + 79d = \frac{47}{4}$$

The 75th term is

$$a + 74d = 14.5 = \frac{29}{2}$$

Double it:

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Subtract:

$$69d = 29 - \frac{47}{4} = \frac{69}{4}$$

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$$171 = \frac{X(X-33)}{8}$$

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$$X = \frac{33+81}{2} = 57$$

FINAL ANSWER

$X = 57$

PAST PAPER QUESTION Q#: 21 Marks: 2

TOPIC: **Transformations of Graphs**

May 2023 P1H - May 2023 | P1H

21 A curve has equation $y = f(x)$

There is only one turning point on the curve.
The coordinates of this turning point are (6, 5)

Write down the coordinates of the turning point on the curve with equation

(a) $y = f(x - 4)$

(.......................... , ...)

(1)

(b) $y = f(3x)$

(.......................... , ...)

(1)

(Total for Question 21 is 2 marks)

PAST PAPER SOLUTION

Q#: 21 Marks: 2

TOPIC: **Transformations of Graphs**

May 2023 P1H - May 2023 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

The turning point of $y = f(x)$ is $(6, 5)$.
For

$$y = f(x - 4)$$

the graph moves 4 units right:

$$(6, 5) \to (10, 5)$$

For

$$y = f(3x)$$

the x-coordinate is divided by 3:

$$(6, 5) \to (2, 5)$$

FINAL ANSWER

$(10, 5)$ and $(2, 5)$

PAST PAPER QUESTION Q#: 21 Marks: 2

TOPIC: **Transformations of Graphs**

May 2023 P1H - May 2023 | P1H

21 A curve has equation $y = \mathrm{f}(x)$

There is only one turning point on the curve.
The coordinates of this turning point are (6, 5)

Write down the coordinates of the turning point on the curve with equation

(a) $y = \mathrm{f}(x - 4)$

(.......................... , ...)
(1)

(b) $y = \mathrm{f}(3x)$

(.......................... , ...)
(1)

(Total for Question 21 is 2 marks)

PAST PAPER SOLUTION

Q#: 21 Marks: 2

TOPIC: **Transformations of Graphs**

May 2023 P1H - May 2023 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

The turning point of $y = f(x)$ is $(6, 5)$.
For

$$y = f(x - 4)$$

the graph moves 4 units right:

$$(6, 5) \to (10, 5)$$

For

$$y = f(3x)$$

the x-coordinate is divided by 3:

$$(6, 5) \to (2, 5)$$

FINAL ANSWER
$(10, 5)$ and $(2, 5)$

PAST PAPER QUESTION

Q#: 22 Marks: 6

TOPIC: **Circles, Arcs & Sectors**

May 2023 P1H - May 2023 | P1H

22 The diagram shows two circles with centre O and a regular pentagon $ABCDE$

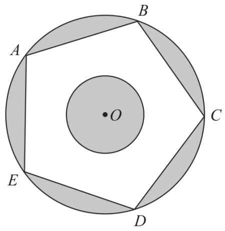

Diagram **NOT** accurately drawn

A, B, C, D and E are points on the larger circle.
The pentagon has sides of length 8 cm.

The diagram is shaded such that

shaded area = unshaded area

Work out the radius of the smaller circle.
Give your answer correct to 3 significant figures.

.... cm

(Total for Question 22 is 6 marks)

PAST PAPER SOLUTION

Q#: 22 Marks: 6

TOPIC: **Circles, Arcs & Sectors**

May 2023 P1H - May 2023 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Moved to Circles, Arcs & Sectors.

Method

The regular pentagon has side length 8 cm.
For a regular pentagon,

$$R = \frac{s}{2\sin 36^\circ}$$

where R is the radius of the larger circle.

$$R = \frac{8}{2\sin 36^\circ} = 6.8052\ldots$$

Area of the larger circle:

$$\pi R^2 = 145.4898\ldots$$

Area of the regular pentagon:

$$\frac{5s^2}{4\tan 36^\circ} = \frac{5(8)^2}{4\tan 36^\circ} = 110.1106\ldots$$

Let the radius of the smaller circle be r.
The shaded area is:

$$\text{larger circle area} - \text{pentagon area} + \pi r^2$$

The unshaded area is:

$$\text{pentagon area} - \pi r^2$$

These are equal, so

$$145.4898 - 110.1106 + \pi r^2 = 110.1106 - \pi r^2$$

$$2\pi r^2 = 2(110.1106) - 145.4898$$

$$\pi r^2 = 37.3657\ldots$$

$$r = \sqrt{\frac{37.3657\ldots}{\pi}} = 3.4487\ldots$$

FINAL ANSWER

$\boxed{3.45 \text{ cm}}$

PAST PAPER QUESTION

Q#: 22 Marks: 6

TOPIC: **Circles, Arcs & Sectors**

May 2023 P1H - May 2023 | P1H

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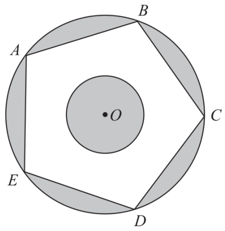

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The pentagon has sides of length 8 cm.

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shaded area = unshaded area

Work out the radius of the smaller circle.
Give your answer correct to 3 significant figures.

.... cm

(Total for Question 22 is 6 marks)

PAST PAPER SOLUTION

Q#: 22 Marks: 6

TOPIC: **Circles, Arcs & Sectors**

May 2023 P1H - May 2023 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Moved to Circles, Arcs & Sectors.

Method

The regular pentagon has side length 8 cm.
For a regular pentagon,

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$$\pi r^2 = 37.3657\ldots$$

$$r = \sqrt{\frac{37.3657\ldots}{\pi}} = 3.4487\ldots$$

FINAL ANSWER

$\boxed{3.45 \text{ cm}}$

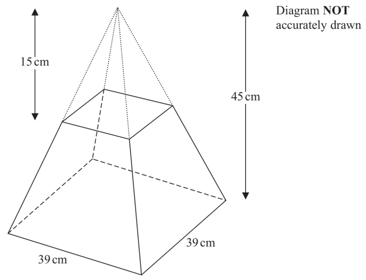
PAST PAPER QUESTION

Q#: 23 Marks: 5

TOPIC: **Area & Volume of Similar Shapes**

May 2023 P1H - May 2023 | P1H

23 A frustum is made by removing a small square-based pyramid from a similar large squared-based pyramid as shown in the diagram.



The height of the small pyramid is 15 cm.
The height of the large pyramid is 45 cm.
The square base of the large pyramid has side length 39 cm.

Work out the **total** surface area of the frustum.
Give your answer correct to the nearest whole number.

.... cm^2

(Total for Question 23 is 5 marks)

PAST PAPER SOLUTION

Q#: 23 Marks: 5

TOPIC: **Area & Volume of Similar Shapes**

May 2023 P1H - May 2023 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

The small and large square-based pyramids are similar.
Their heights are

$$15 : 45 = 1 : 3$$

So the side length of the top square of the frustum is

$$\frac{39}{3} = 13 \text{ cm}$$

The vertical height of the frustum is

$$45 - 15 = 30 \text{ cm}$$

For one trapezium face, the difference between the half-side lengths is

$$\frac{39 - 13}{2} = 13 \text{ cm}$$

So the slant height of the trapezium face is

$$\sqrt{30^2 + 13^2} = \sqrt{1069}$$

Area of one trapezium face:

$$\frac{1}{2}(39 + 13)\sqrt{1069} = 26\sqrt{1069}$$

Area of four trapezium faces:

$$4(26\sqrt{1069}) = 104\sqrt{1069}$$

Add the top and bottom square faces:

$$13^2 + 39^2 = 169 + 1521 = 1690$$

Total surface area:

$$1690 + 104\sqrt{1069} = 5090.338\ldots$$

Correct to the nearest whole number:

$$5090$$

FINAL ANSWER

$\boxed{5090 \text{ cm}^2}$

PAST PAPER QUESTION

Q#: 23 Marks: 5

TOPIC: **Area & Volume of Similar Shapes**

May 2023 P1H - May 2023 | P1H

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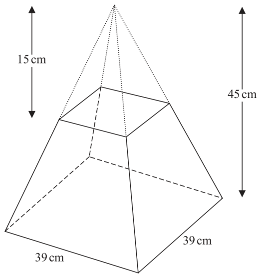

Diagram **NOT** accurately drawn

The height of the small pyramid is 15 cm.
The height of the large pyramid is 45 cm.
The square base of the large pyramid has side length 39 cm.

Work out the **total** surface area of the frustum.
Give your answer correct to the nearest whole number.

.... cm^2

(Total for Question 23 is 5 marks)

PAST PAPER SOLUTION

Q#: 23 Marks: 5

TOPIC: **Area & Volume of Similar Shapes**

May 2023 P1H - May 2023 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

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FINAL ANSWER

$\boxed{5090 \text{ cm}^2}$